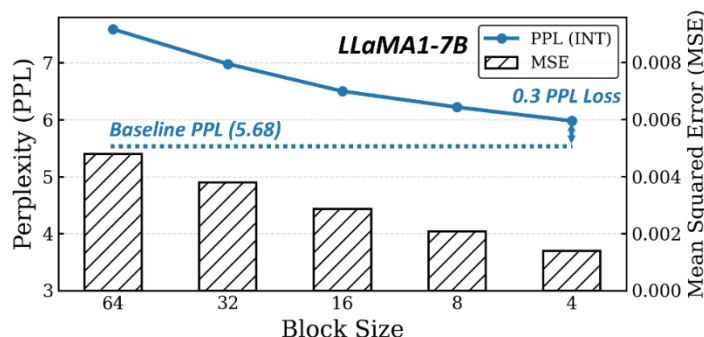
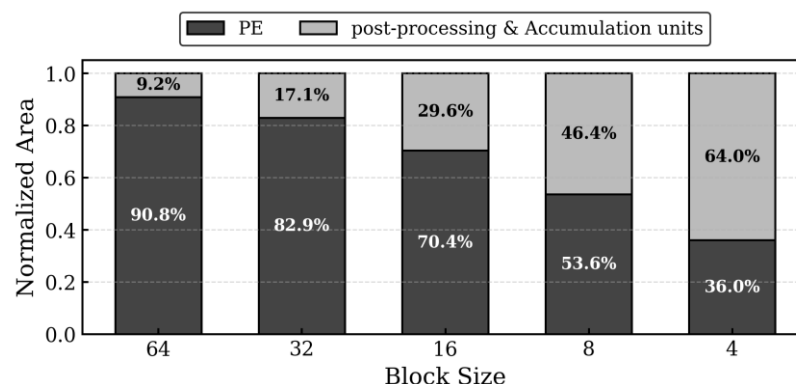


A GEMM Accelerator for Hardware-Efficient LLM Inference via Outlier-Aware Merge-and-Split Block Quantization (1)

- ◆ **Goal:** To achieve scalable and energy-efficient LLM inference via a **co-designed GEMM accelerator** with outlier-aware **merge-and-split (M&SB) quantization**
- ◆ **Motivation:** Reducing block size uniformly improves accuracy but incurs **fragmentation overhead and linear hardware cost**; **adaptive methods** improve flexibility yet still suffer from **poor scalability and suboptimal adaptation**
- ◆ **Limitations of Block Quantization**
 - **Limitation 1 (Conventional)**
 - Uniform block sizes ignore outlier distribution, causing **unnecessary fragmentation and metadata overhead**
 - **Limitation 2 (Adaptive)**
 - Adaptive method suffers from poor scalability, as smaller blocks require **more post-processing units (PPUs) and accumulators**
 - **Limitation 3 (Adaptive)**
 - MSE-based selection is too costly for online use, **while simplified adaptive criteria lead to suboptimal accuracy**

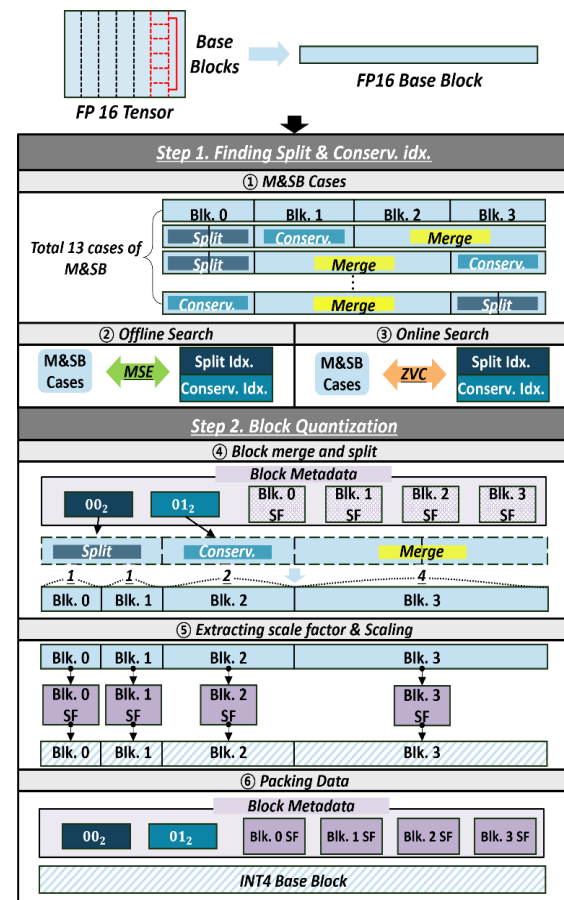


<Effect of block size on perplexity and MSE>



<Normalized area breakdown of PE, post-processing, and accumulation units under different block sizes>

- ◆ **Goal:** To achieve scalable and energy-efficient LLM inference via a **co-designed GEMM accelerator** with outlier-aware **merge-and-split (M&SB) quantization**
- ◆ **Solution/Contribution:** **M&SB with ZVC adaptively splits and merges blocks using zero-counting** for efficient outlier-aware quantization
- **1) merge-and-split block (M&SB) block quantization:**
 - **Generate candidate block configurations via selective split and merge**
 - Partition each base block into four sub-blocks
 - **Split one sub-block and merge two non-outlier sub-blocks**
 - Store split/conservation indices and FP8 scale factors as metadata
 - **Reduce fragmentation overhead while improving accuracy**
- **2) zero-counting and variance-based criterion (ZVC):**
 - **Select configuration among M&SB candidates using zero-counting**
 - Apply MSE-based selection for weights, and ZVC for activations
 - **Select the configuration with the minimum number of quantized zeros**
 - Use variance only as a tiebreaker
 - **Lightweight alternative to MSE**



< Framework of M&SB quantization >

A GEMM Accelerator for Hardware-Efficient LLM Inference via Outlier-Aware Merge-and-Split Block Quantization (3)

- ◆ **Goal:** To achieve scalable and energy-efficient LLM inference via a **co-designed GEMM accelerator** with outlier-aware **merge-and-split (M&SB) quantization**
- ◆ **Solution/Contribution:** Propose BGM, a GEMM accelerator with a dedicated **Block Processing Unit (BPU)** supporting mixed block sizes without linear hardware cost growth

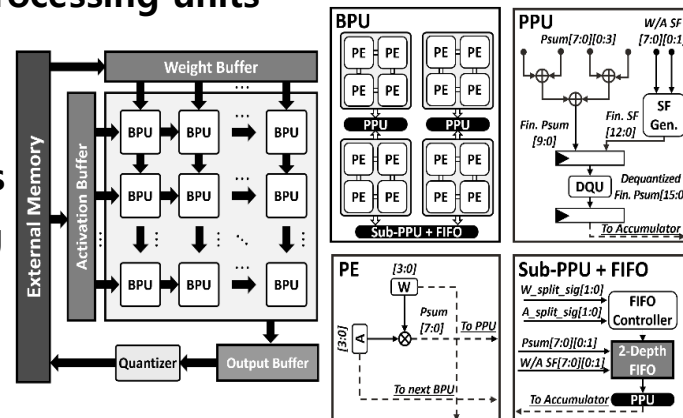
3) Architecture of BGM:

- BGM adopts a systolic array architecture with BPU-based processing units

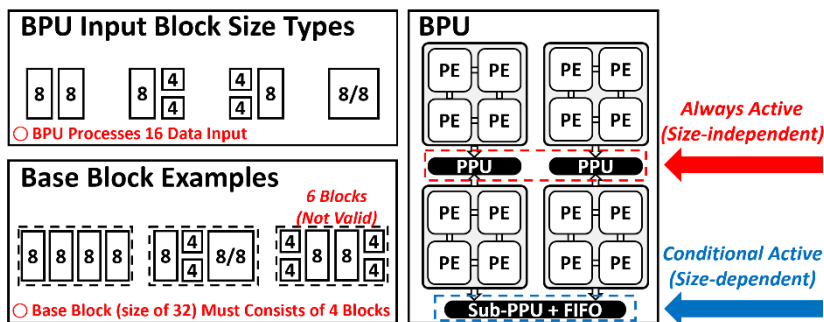
- Each BPU supports mixed block sizes from M&SB quantization
- PPUs consist of dequantization, scaling, and adder trees
- Unified processing for heterogeneous block structures

- BPU processes heterogeneous block sizes using shared PPU

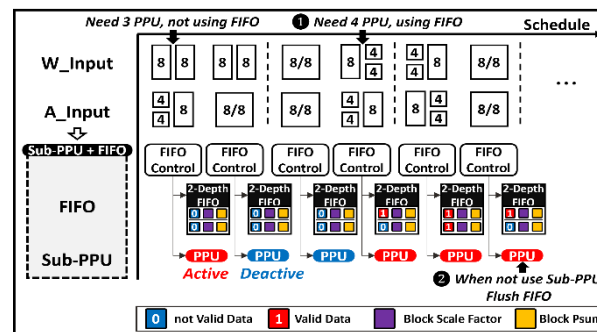
- Sub-PPU + FIFO-based dynamic scheduling
- PPUs are activated conditionally based on input block size
- Avoid linear hardware cost growth



<Overview of BGM architecture>



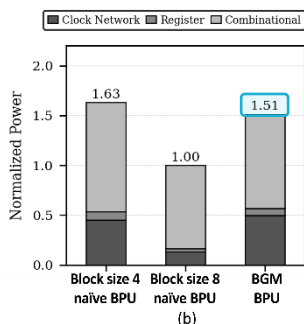
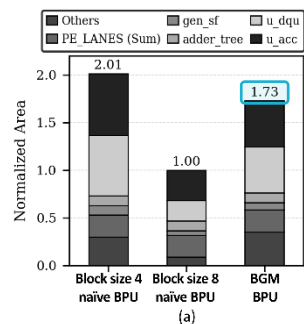
<BPU input configurations>



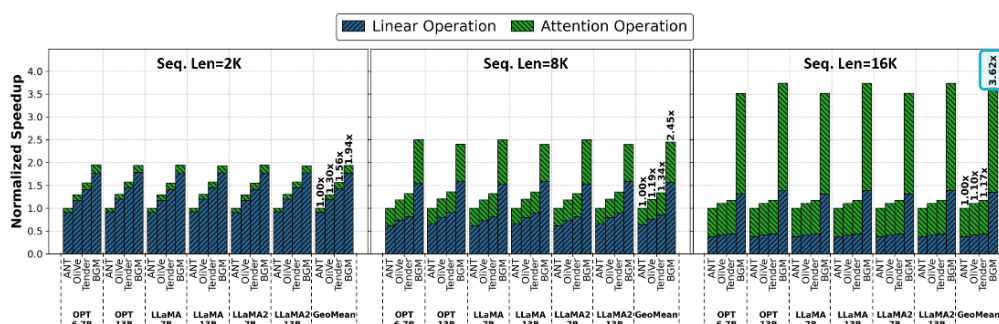
<Sub-PPU and FIFO scheduling example>

Experimental Results

- BGM achieves **28.39% area reduction** and **11.80% power savings** over naive designs by leveraging shared PPUs and eliminating redundant accumulators
- It achieves **2.67× average speedup** (up to **3.62×**) and **2.50× energy reduction** (up to **3.46×**) over ANT (representative baseline)
- BGM outperforms prior adaptive methods, achieves **near next-step block size accuracy**, and **surpasses MSE-based quantization**



<Area (a) and power (b) comparison of the BPU against naive designs>

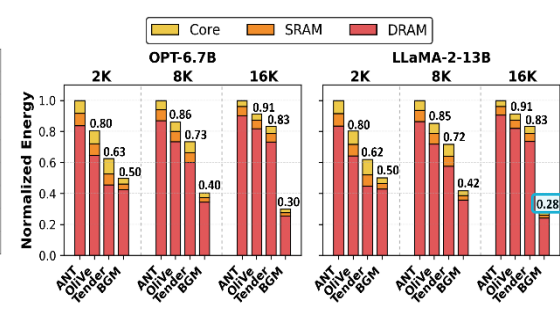


<Normalized speedup on various accelerators>

Method	Effective Block Size	Data Format	Linear (W/A)	Atten. (KV/A)	OPT 6.7B	OPT 13B	LLaMA1 7B	LLaMA1 13B	LLaMA2 7B	LLaMA2 13B
Baseline	-	FP16	16/16	16/16	10.86	10.13	5.68	5.09	5.47	4.88
ANT [25]	-	Mixed	4/4	16/16	6.4E+3	9.6E+3	61.35	53.74	124.18	333.29
OliVe [26]	-	INT	4/4	16/16	39.18	65.42	32.15	15.64	44.07	50.28
Tender [27]	-	INT	4/4	16/16	13.56	16.43	23.85	13.68	36.47	55.08
Quarot [45]	-	INT	4/4	4/16	-	-	6.34	5.58	6.10	5.40
ATOM [46]	64	INT	4/4	4/16	-	-	6.16	5.46	6.12	5.31
MXFP [28]	32	FP	4/4	4/4	22.51	12.88	10.20	7.80	11.18	6.98
INT-sym	32	INT	4/4	4/4	11.90	11.45	6.98	5.95	6.90	5.96
AMXFP [29]	16	FP	4/4	4/4	13.06	11.90	6.25	5.52	6.22	5.47
BOLD-Q [32]	16	LNS	4/4	4/4	11.45	10.97	6.18	5.45	6.07	5.37
BlockDialect [30]	14.2	FXP	4/4	4/4	11.45	-	6.16	-	6.08	-
BGM (ours)	14.2	INT	4/4	4/4	11.37	10.75	6.16	5.42	6.07	5.30

Entries with unreported or unreproducible results are marked “-”.

<Quantization Results for Various LLMs on WikiText-2 (perplexity ▼)>



<Normalized energy on various accelerators>

Method	Eff. Blk. Size	OPT 6.7B	OPT 13B	LLaMA1 7B	LLaMA1 13B	LLaMA2 7B	LLaMA2 13B	Mean Δ PPL
Baseline	-	10.86	10.13	5.68	5.09	5.47	4.88	-
INT-sym	32	11.90	11.45	6.98	5.95	6.90	5.96	1.17
BGM (MSE)	28.4	11.58	11.18	6.57	5.61	6.42	5.54	0.80
BGM (ours)	28.4	11.57	11.16	6.46	5.62	6.41	5.56	0.78
INT-sym	16	11.55	10.98	6.50	5.64	6.38	5.59	0.76
BGM (MSE)	14.2	11.32	10.76	6.26	5.43	6.01	5.31	0.50
BGM (ours)	14.2	11.37	10.75	6.16	5.42	6.07	5.30	0.49
INT-sym	8	11.31	10.79	6.22	5.42	6.04	5.33	0.50
BGM (MSE)	7.1	11.24	10.64	6.05	5.30	5.86	5.18	0.36
BGM (ours)	7.1	11.27	10.62	5.98	5.29	5.83	5.15	0.34
INT-sym	4	11.17	10.53	5.98	5.27	5.80	5.14	0.30
BGM (MSE)	3.6	11.13	10.54	5.87	5.21	5.69	5.06	0.23
BGM (ours)	3.6	11.15	10.56	5.86	5.20	5.68	5.04	0.23

<WikiText-2 perplexity under full INT4 quantization (W/A/KV) with ZVC and MSE-based M&SB (↓)>